

CHAPTER 2

Preferences

Chapter Summary

Investors generally dislike the risk of losses during bad times. Optimal portfolio choice trades off these risks with the potential of returns. While mean-variance utility treats gains and losses symmetrically, other models of preferences allow investors to seek safety first, to weight the pain of losses more heavily than the benefit of gains, and to have their utility depend on their past consumption (habit) and the returns or actions of other investors.

1. Picking Up Nickels and Dimes in Front of a Steamroller

Short volatility strategies are like the sly fox in the tale of the little gingerbread man. Just as the fox hoodwinked the tasty gingerbread man to ride on his back to cross a river, the steady and high profits generated by short volatility strategies during stable times often lure investors into complacency. Many investors pile into short volatility strategies after they have seen stable profits for many years. Then, when volatility spikes during market crashes, short volatility strategies suddenly turn around and give investors a nasty bite. Most of the profits earned during normal times are given back at a quick snap.

Figure 2.1 graphs the cumulated wealth of \$1 invested in a short volatility strategy at the end of March 1989 to December 2011.¹ Volatility strategies are implemented in derivatives markets using options, variance swaps, and even derivatives on derivatives such as options on swaps. (Merrill Lynch's index is produced by trading variance swaps.) Volatility traders also buy and sell securities with embedded options such as convertible notes and bonds.

The returns graphed in Figure 2.1 are for an investor who has sold volatility—this investor collects a premium during normal times for selling volatility protection. On the other side, the people paying the premiums have purchased

¹ Monthly returns on the volatility strategy are given by the MLHFEV1 index produced by Merrill Lynch.